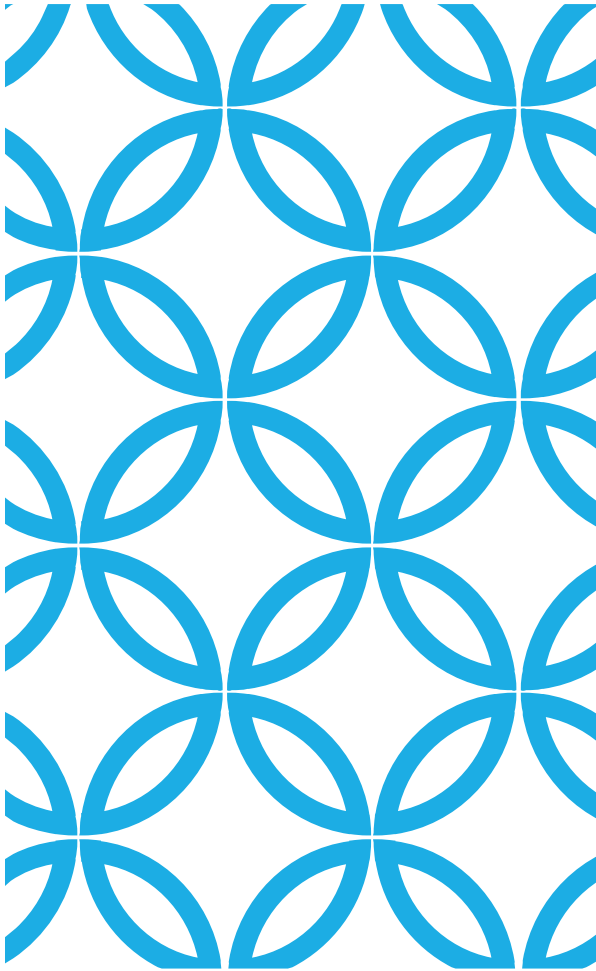
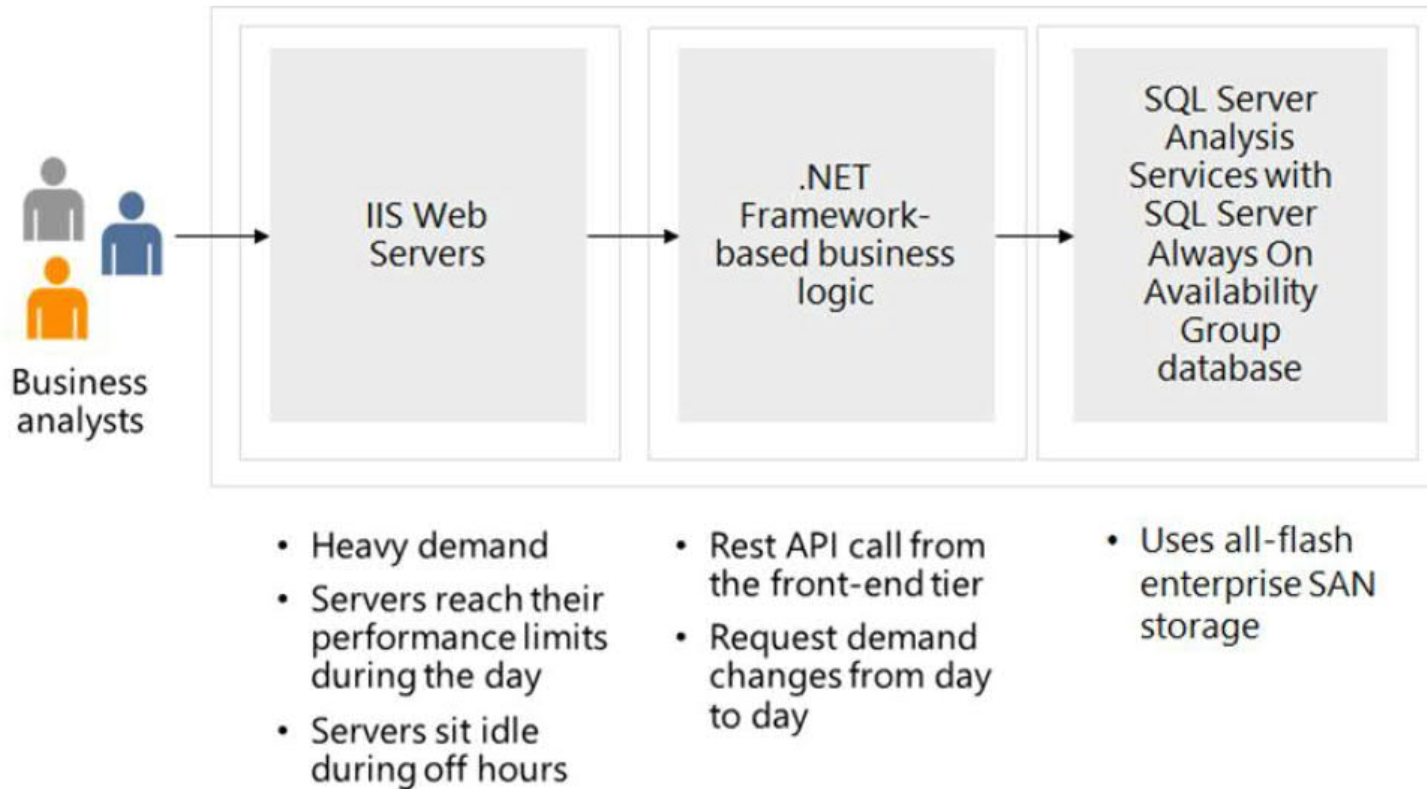




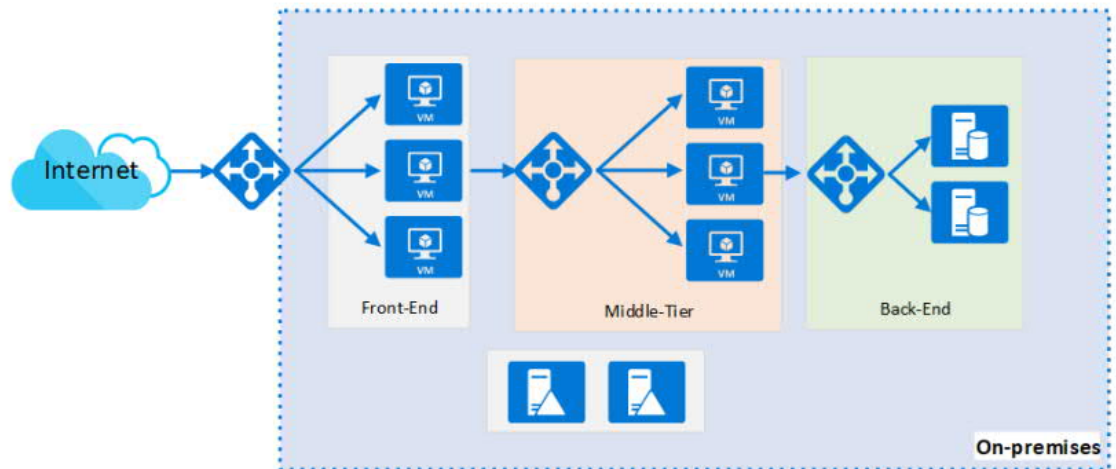
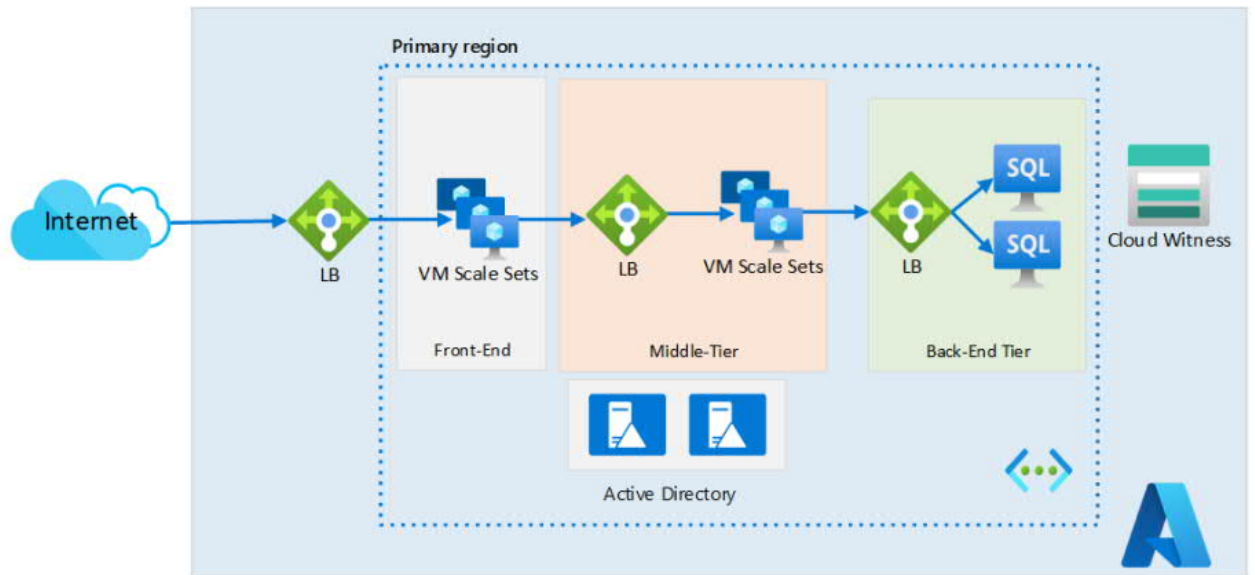
CLOUD MIGRATION AND DESIGN - CASE STUDIES

CS 4037 – Introduction to Cloud Computing
FAST-NUCES

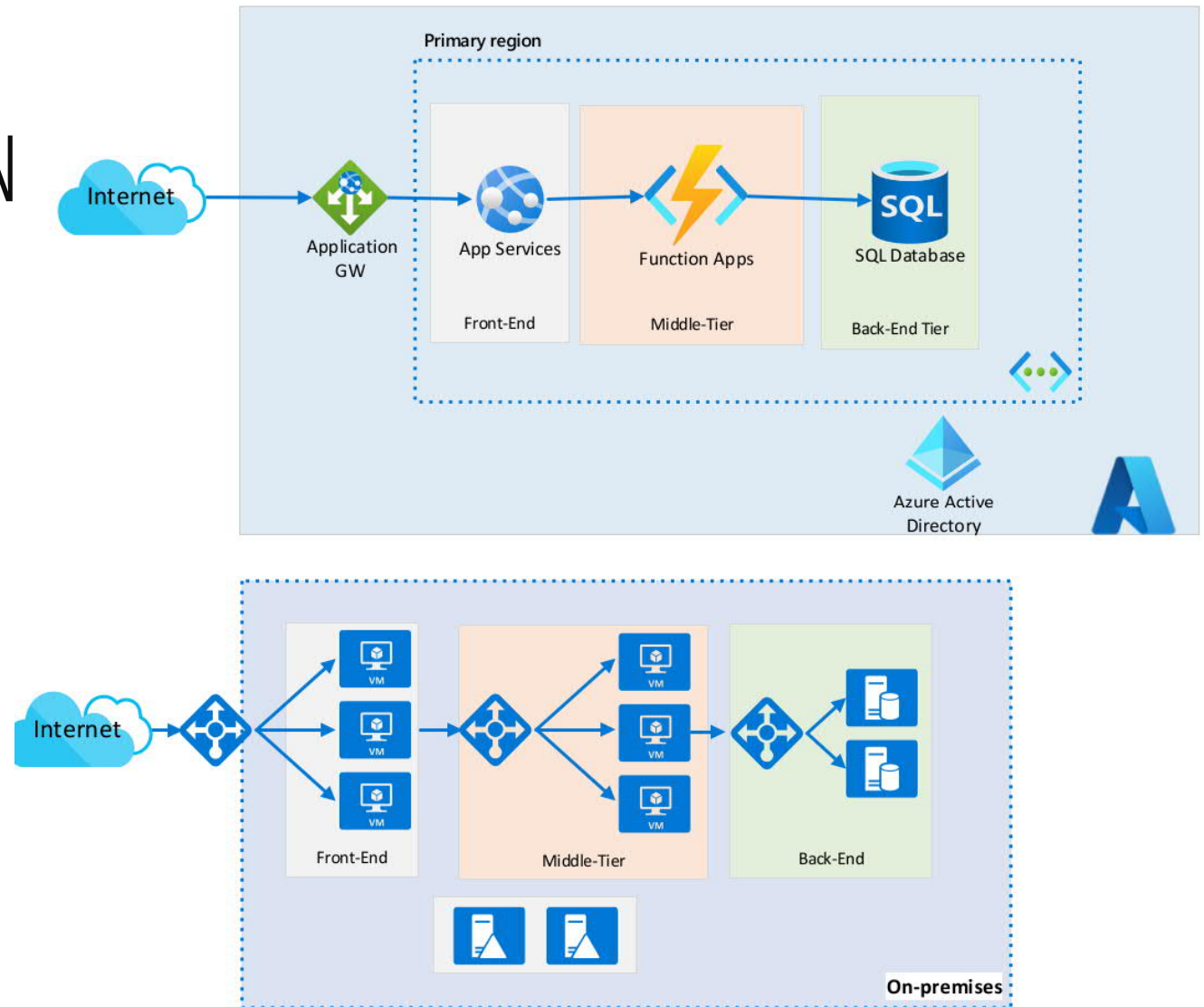
DESIGN COMPUTE CASE



IAAS SOLUTION



PAAS SOLUTION



DESIGN NON-RELATIONAL STORAGE CASE STUDY

Requirements:

Tailwind Traders wants to reduce storage costs by reducing duplicate content and, whenever applicable, migrating it to the cloud.

They would like a solution that centralizes maintenance while still providing world-wide access for customers who marketing literature.

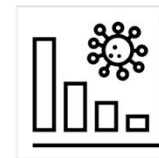
Also, they would like the external vendor to work on their media files and update as necessary.

Additionally, they would like to address the storage of company data files.



Media files

- Product photos and feature videos
- JPEG and MP4 are most common formats



Marketing literature

- Customer stories, sales flyers, sizing charts, and eco-friendly manufacturing information
- PDF format is the most common



Corporate documents

- Internal documents – some sensitive
- Mostly Office formats like Word and Excel

DESIGN NON- RELATIONAL STORAGE CASE STUDY

Blob storage is recommended for the media and corporate files. Blob storage is less expensive, offers the immutable storage requirements for legal, and supports API access for internal applications.

Azure Files is recommended for the marketing literature. Azure Files is needed for marketing since the files will be accessed via SMB internally.

Marketing literature access latency for internal users could be reduced by using a local Windows Server and File Sync



Zone redundant storage is recommended. An argument could be made for Geo-redundant zone storage if the files are mission critical.

Read access geo-redundant storage could be used if the front end could make use of a secondary region.

This decision would depend on the customer's locations and traffic.

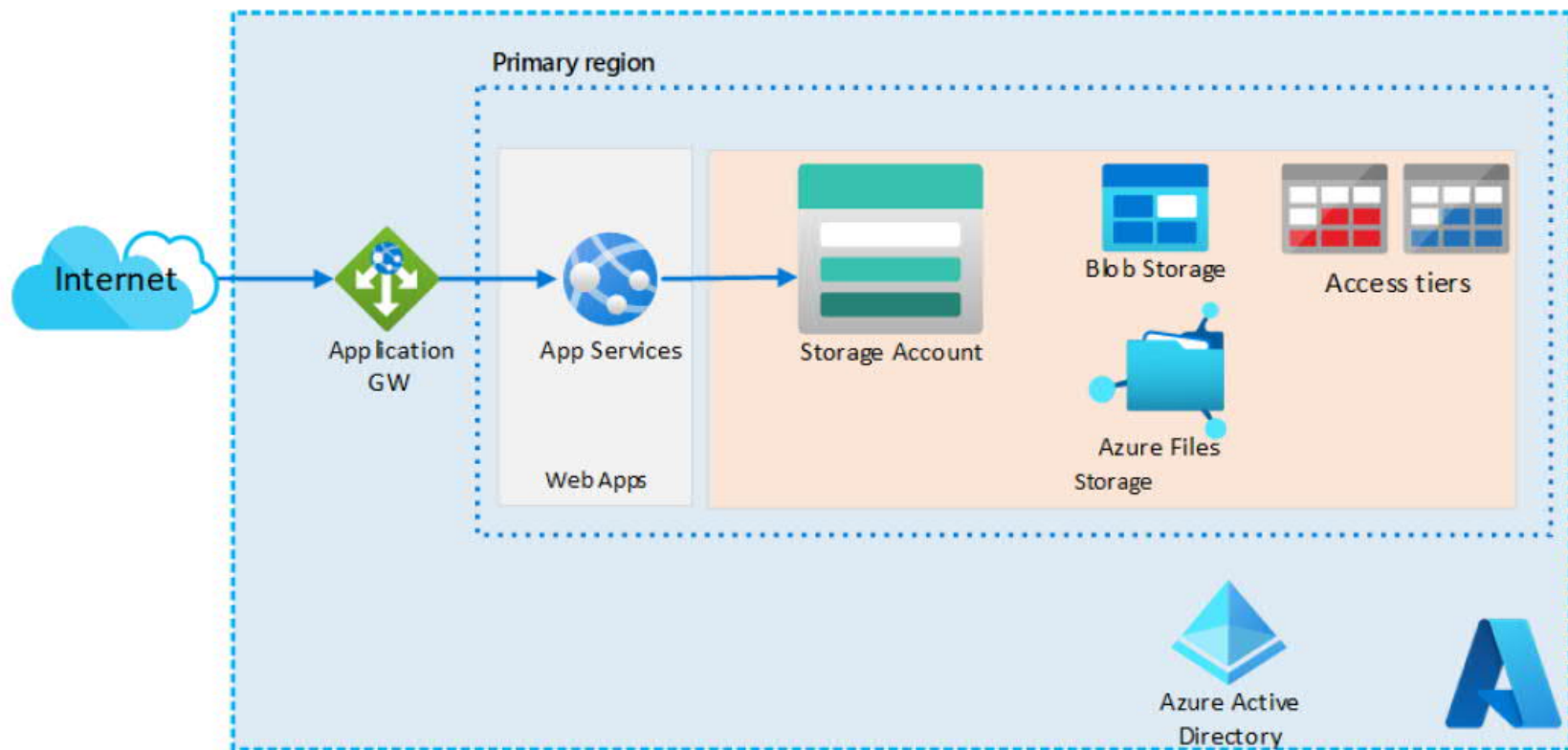


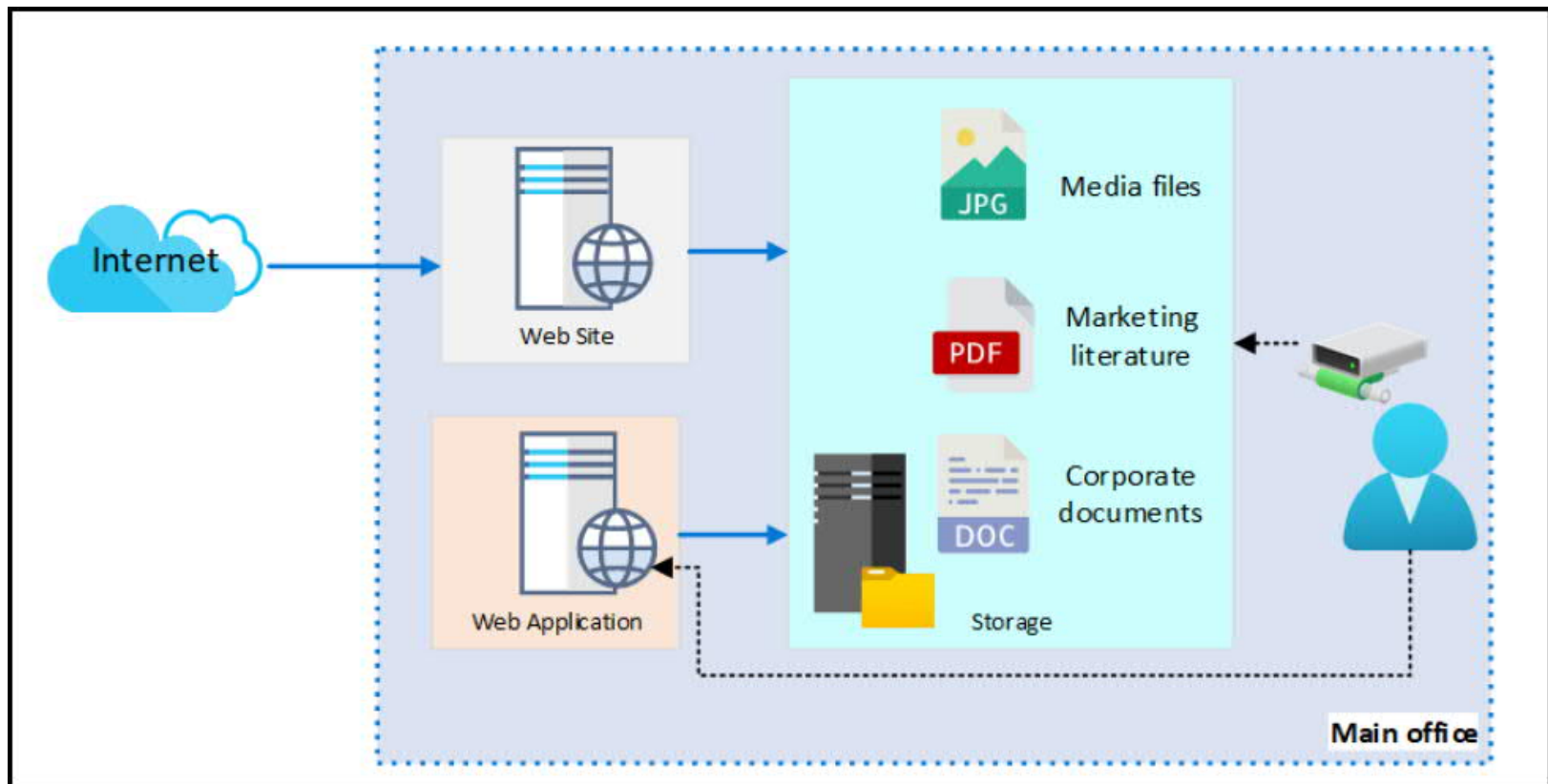
The hot tier should be used for all media, marketing literature and corporate documents less than one year old. The archive tier or cold tier should be used for corporate documents older than one year.

This decision is based on retrieval latency, storage duration and a desire to reduce costs.

Discuss that there is not enough information to decide between Cold and Archive and ask what other questions they may want to ask Tailwind Traders.

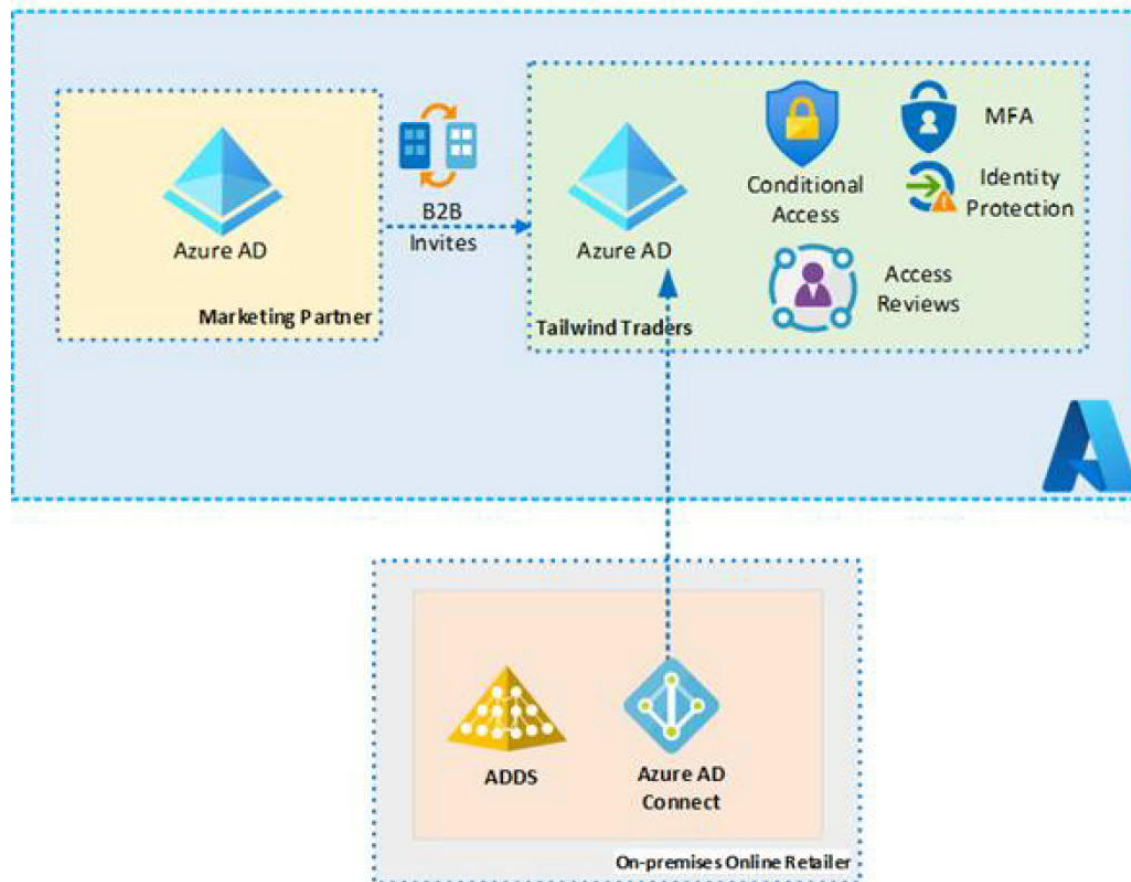
Lifecycle management should be used to convert corporate documents to a cheaper storage tier after one year.

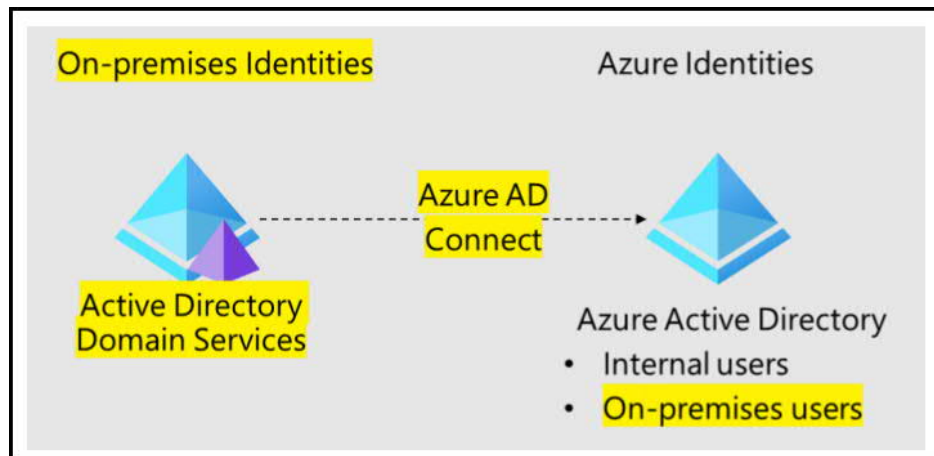




DESIGN AUTHENTICATION AND AUTHORIZATION SOLUTIONS

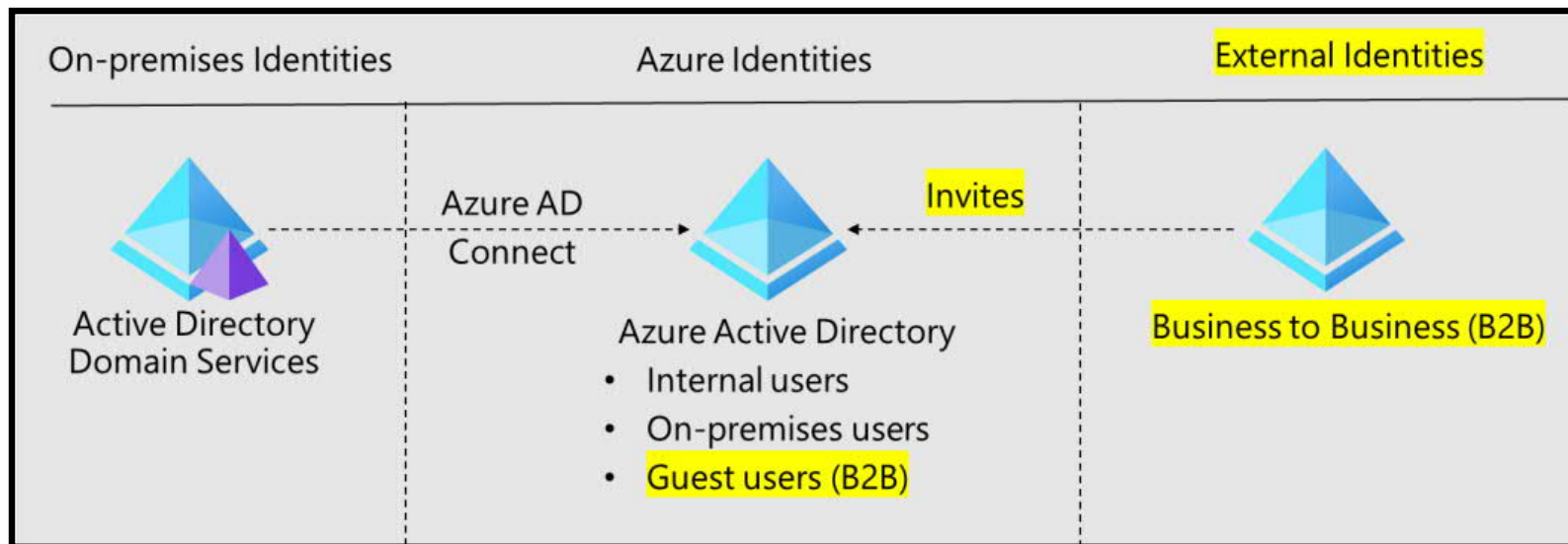
1. New employee user accounts.





Note:

Azure AD Connect is now called Entra ID Connect
Azure Active D is now called Entra ID.



NEW IDENTITY SOLUTION FEATURES

- Use MFA for privileged roles like administrators. Consider MFA for the partner accounts.
- Use access reviews to ensure users changing jobs still have the correct permissions.
- Use RBAC to ensure permissions are correct. Design at the group level.
- Require users to access applications only from managed devices.
- Block access from untrusted sources, such as access from unknown or unexpected locations.
- Establish user and sign-in risk policies.